

Speed Ripples Reduction of PMSM Drives under Low-Speed Working Conditions

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Abstract- The present work deal with the undesired speed ripples caused by parasitic torques at low-speed conditions of Permanent Magnet Synchronous Machine (PMSM) based drives. The proposed speed ripples reduction technique is based on a power harmonic compensator where the damping signals are superposed to the reference current in a Field Oriented Control (FOC) scheme. The design procedure of the torque ripple compensator takes into account the local stability of the system. The practical feasibility and effectiveness of the proposed approach are verified under a small-scale laboratory test bench. Furthermore, the impact of the proposed method on the machine vibration is measured.

Index Terms—Permanent magnet synchronous machine, speed ripples reduction, low-speed operation, P-Q theory.

NOMENCLATURE

T_e	Electromagnetic torque
T_L	Load torque.
ψ_f	Flux linkage
i_d, i_q	dq -axis currents.
v_d, v_q	dq -axis input voltages.
ω	Electrical speed.
k_{com}	Compensation gain.

Subscripts

d, q	Park frame.
ref	Reference.
$Comp$	Compensation

I. INTRODUCTION

Thank's to their efficiency and high power density Permanent magnet synchronous motors (PMSM) become the most used technologies for transportation electrification [1] [2]. However, one of the main drawbacks of PMSM drives is the occurrence of parasitic torque pulsations at low-speeds which lead to undesired speed oscillations [3] [4]. These low-frequency oscillations are particularly undesirable in challenging applications

such as motion control machines (machine tool designs) and electromobility applications. Furthermore, torque ripple can cause stress on mechanical parts resulting in premature aging of some parts, produce acoustic noise and, which can affect user comfort in vehicular applications. At higher operating speeds, the pulsation frequencies are naturally filtered out by the mechanics (rotor and load inertia) [5] [6] [7].

Torque oscillations at low speeds can be caused by different factors which are due to:

- Internal factors: imperfections of the machine induce parasitic torques. The imperfections are related to the rotor structure and distribution of the stator teeth, which are essentially translated into parasitic torques: cogging torque and one related to magnetic harmonics (flux harmonics). The cogging torque, due to the stator slots, results harmonics ranges at " $N_s f_e$ ". N_s is the number of slots per pole pair and f_e is the electrical shaft speed in Hz. In addition, flux harmonics consist mainly on impairs harmonics 5, 7, 11 and this result a parasitic torques multiples of " $6f_e$ ".
- External factors: it present harmonics induced mainly by the load and its mechanical coupling, harmonics and delays produced by the converter-control unit as well as measurement conditioning problems (delays, offsets, ... etc.). The resulted harmonics of some issues such as coupling load can lead to lower harmonics then the fundamental frequency and is more difficult to reject it by classical controllers.

Many control techniques strategies and methods have been proposed in order to reduce the torque ripple in the PMSM [8] [9]. Among the most investigated active strategies to address this problem, the iterative learning controls (ILC) have been proposed in several works for the reduction of torque oscillations in low speed operation [10] [11].

This approach has two advantages: i) from a theoretical point of view, this solution allows to produce an infinite gain and reject in a total way the periodic disturbances; ii) the iterative algorithm is simple to implement and its robustness performances can be preserved on a large frequency range. Besides these advantages, these kinds of control engender excessive stress on the power electronic components and the gains of the ILC must be limited to avoid instability problems. Other studies [12] [13] are based on the assumption of a prior knowledge of the harmonics coming from the parasitic couples, and propose control strategies designed to reject these specific harmonics. These strategies are based on the proportional integral-resonant corrector (PIR) which allows producing important gains on selected harmonic ranks particularly two and multiple of six [14] [15] [16] [17].

In this paper, an active speed-harmonic compensator based on a P-Q theory is proposed to ensure speed smoothness at low speed operation of PMSM motors. The approach is based on the use of the motor power calculation to find the suitable compensation current to be injected in a conventional field oriented control. The idea of the use of a P-Q theory to reduce torque ripple in PMSM is inspired in literature works about the active filters. In the studied case, PMSM, the use of PQ-theory as active speed-harmonic compensator allows:

- In first step to identify the current disturbance which is outside of the PMSM controller bandwidth ;
- In second step to modify the main reference current in such a way that the undesired disturbances are rejected.

The design procedure of the proposed torque-harmonic compensator takes into account the local stability of the system.

The reminder of this paper is presented as follows. A description of the studied system and its control are presented in section II. In section III, the design principle of the proposed compensation method is provided. Section IV presents simulation results. In section V, experimental results validate the effectiveness of the proposed compensation technique. Finally, section VI underlines the contributions of this work.

II. SYSTEM DESCRIPTION AND CONTROL

The modeling of the PMSM and the control laws are presented below.

A. Modeling

This study considers the case of smooth poles PMSM and for the modeling the following hypothesis are considered: PMSM is unsaturated and eddy currents and hysteresis losses are neglected. The electrical equations of the PMSM in the Park frame are expressed as follows:

$$v_d = R_s i_d + L_d \left(\frac{di_d}{dt} \right) - \omega L_q i_q \quad (1)$$

$$v_q = R_s i_q + L_q \left(\frac{di_q}{dt} \right) + \omega L_d i_d + \omega \psi_f \quad (2)$$

where v_d, v_q are the d-axis and q-axis stator voltage. R_s is the stator resistance, L_d, L_q are the d-axis and q-axis inductances. ψ_f is the permanent magnet flux and ω is the electrical rotor speed. i_d, i_q are the d-axis and q-axis stator current. The PMSM parameters are given in the Appendix.

The mechanical equation is expressed as:

$$J \frac{d\Omega}{dt} = T_e - T_L - f\Omega \quad (3)$$

where J is the inertia of the rotor Ω is the mechanical rotor speed. T_L is the load torque, f is the viscous frictional coefficient, and the electromagnetic torque T_e has the following equation:

$$T_e = p(\psi_f - (L_q - L_d)i_d)i_q \quad (4)$$

where p is motor pair of poles with $\omega = p\Omega$ and ψ_f is the permanent magnet flux. The torque expression can be simplified to (5) when considering the case of smooth poles machine ($L_q = L_d = L_s$).

$$T_e = p\psi_f i_q \quad (5)$$

The diagram of the PMSM in dq-frame and its control are presented in Fig. 1. The proposed compensator effort noted $i_{q_{Comp}}$ is added to the reference quadratic current i_q^{ref} .

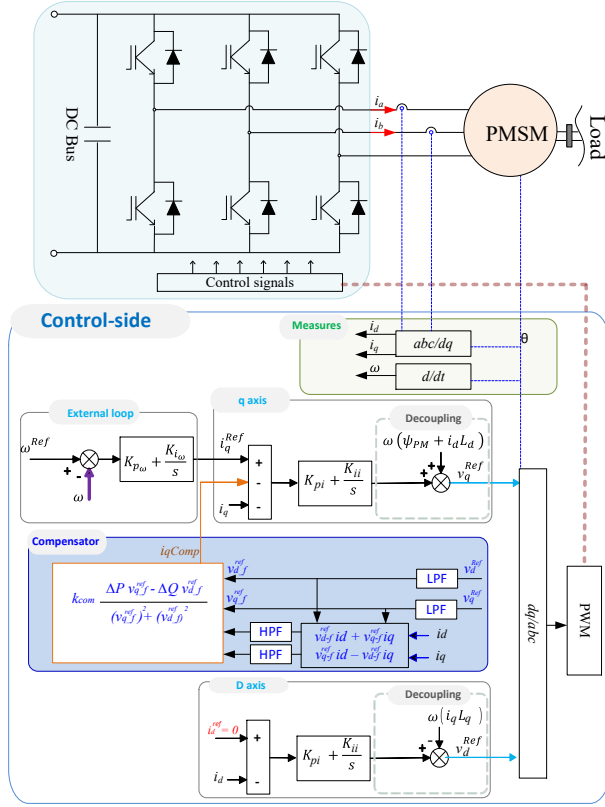


Fig. 1. Scheme of the studied system with its control and the proposed speed ripple compensator.

B. PMSM Control

A field oriented control, as described in Fig. 1, based on cascaded PI controllers is considered. The outer speed controller calculates the reference quadratic current by using the following equation.

$$v_d^{ref} = K_{pi}(i_d^{ref} - i_d) + K_{ii} \int (i_d^{ref} - i_d) dt - \underbrace{\omega_e L_s i_q}_{\text{Decoupling term}} \quad (6)$$

$$v_q^{ref} = K_{pi}(i_q^{ref} - i_q) + K_{ii} \int (i_q^{ref} - i_q) dt + \underbrace{\omega_e L_s i_d + \omega_e \psi_f}_{\text{Decoupling term}} \quad (7)$$

The inner current controller equation is given in (7). The direct current reference i_d^{ref} is fixed to zero in order to maximize the torque.

$$i_q^{ref} = K_{p\omega} (\omega^{ref} - \omega) + K_{i\omega} \int (\omega^{ref} - \omega) dt \quad (8)$$

The control parameters are set by a pole placement methods and they are fixed to: $K_{pi} = 2 \xi L_s \omega_c - R_s$, $K_{ii} = L_s \omega_c^2$, $K_{p\omega} = (2 \xi J \omega_\omega) / p \psi_f$ and $K_{i\omega} = (J \omega_\omega^2) / (p \psi_f)$. ω_c and ω_ω presents respectively the cutoff pulsations of the current and speed loops.

III. PROPOSED SPEED RIPPLE REDUCTION APPROACH

A. Principle

The proposed technic calculates a compensating quadratic current i_{qComp} which is added to the reference of the quadratic current as illustrated in Fig. 1. The calculation of the compensating effort i_{qComp} is based on three steps:

- 1) Active and reactive power calculation.

$$P = v_{d-f}^{ref} i_d + v_{q-f}^{ref} i_q \quad (9)$$

$$Q = v_{q-f}^{ref} i_d - v_{d-f}^{ref} i_q \quad (10)$$

where v_{d-f}^{ref} and v_{q-f}^{ref} are the filtered control inputs calculated by the PI controllers. A low pass filters denoted LPF in Fig. 1 are used.

- 2) Compensating current calculation: the following equation is used to calculate the compensating current.

$$i_{qComp} = k_{com} \frac{\Delta P v_{q-f}^{ref} - \Delta Q v_{d-f}^{ref}}{(v_{d-f}^{ref})^2 + (v_{q-f}^{ref})^2} \quad (11)$$

where k_{com} is a compensating factor. A high pass filters (HPF in Fig. 1) are used to extract the oscillating powers $(\Delta P = (1 - \frac{\omega_f}{s + \omega_f})P)$ and $(\Delta Q = (1 - \frac{\omega_f}{s + \omega_f})Q)$. ω_f is a pulsating frequency and s is a Laplace operator.

B. Design procedure

The sizing of the compensator parameters is done through a small signal study to evaluate the stability of the system with the proposed power harmonic compensator.

After a linearization of the system with its control and compensation method where each state is replaced by $x = x_0 + \Delta x$, the analysis of the denominator poles of the system transfer function or directly eigenvalues of the resulted Jacobien matrix allows to study the impact of the

compensator parameters on the system stability. Fig. 2 illustrates the behavior of the poles when the compensating factor k_{com} is changed from 0 to -0.89. As it can be seen the system become unstable when the gain is bigger than -0.84 (the real part of the eigenvalues become positive).

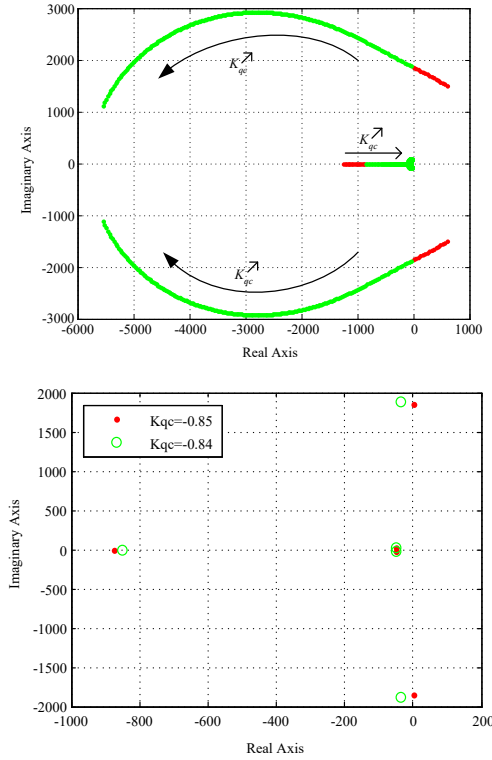


Fig. 2. (a) Poles evolution for $k_{com} = -1 : 0.01 : 1$ with $\omega_r = 2\pi \cdot 10$ rad/s; (stable and unstable poles are presented respectively in green and red). (b) A zoom on a poles evolution for $k_{com} = -0.85 : 0.01 : -0.84$ with $\omega_r = 2\pi \cdot 10$ rad/s; (stable and unstable poles are presented respectively in green and red).

IV. EXPERIMENTAL RESULTS

To verify the effectiveness of the proposed control scheme experiments are carried out with a fast prototyping system (dSPACE DS1103-PPC). The experimental setup, prepared in the IREENA laboratory, Fig. 3, consists on: a PMSM motor controlled through three phase Semikron inverter and a DC generator as brake controlled through a boost converter to ensure the load torque. The test bench is equipped with a DEWE-SIRIUS vibrometer. The PMSM nominal speed is 1500 rpm and the nominal load torque is equal to 16 Nm.

In the following two illustrative tests are shown:

1. Speed ripple ($\Delta\omega = \omega^{ref} - \omega$) under low speed - nominal load torque. For illustration a speed of 30 rpm which present 2% from the nominal speed (1500 rpm) is considered in the following results ;

2. Measure of resulted vibrations without and with the use of the proposed compensator.

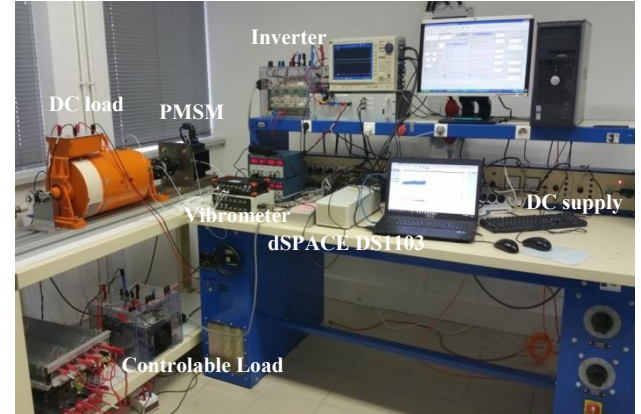


Fig. 3. Overview of the prepared test-bench.

A. Speed Ripples Reduction

Fig. 4 illustrates the temporal behavior of speed ripples ($\Delta\omega = \omega^{ref} - \omega$) without and with the proposed compensator when the reference speed is 30 rpm and the load torque is 16 Nm. It can be seen that the magnitude of speed ripples is effectively decreased when the proposed compensator is activated. The reduction rate is about 50%.

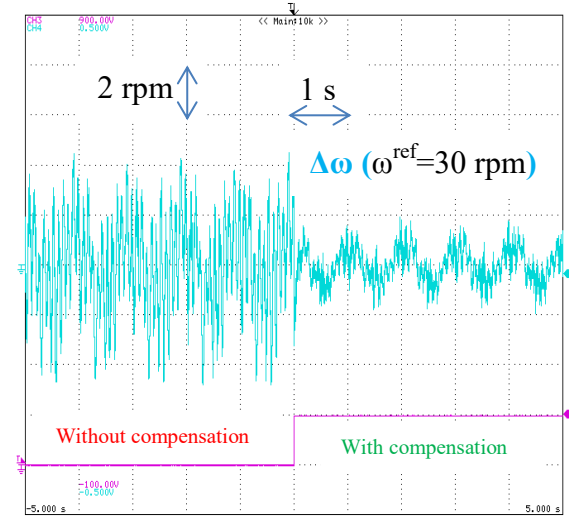


Fig. 4. Speed ripples $\Delta\omega$ under a representative working point (30 rpm – load of 16 Nm) without and with the proposed compensator.

B. Vibration Indicator

To evaluate the influence of the proposed compensation technique on the machine vibrations, the sensors of the vibrometer (DEWE-SIRIUS) shown in Fig. 5 are placed on the machines (motor and load).

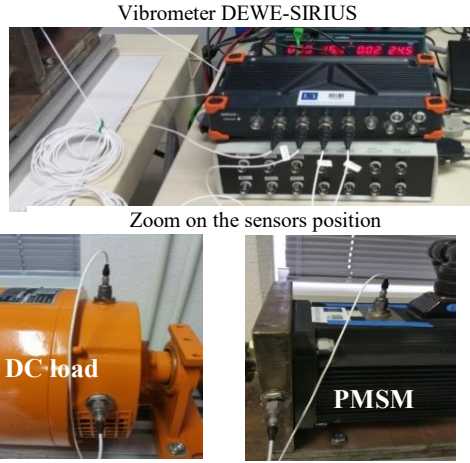


Fig. 5. Vibrometer device and sensors positioning.

Fig. 6 shows the recorded acceleration magnitudes various frequencies respectively without and with the proposed compensator. The operating condition is set to 30 rpm under the nominal torque. These results show important vibration peaks without the compensator (Fig.6 (top)) at specific frequency orders, especially the 6th. The magnitudes of these undesired harmonics are effectively damped when the proposed compensator is activated (Fig. 6(bottom)).

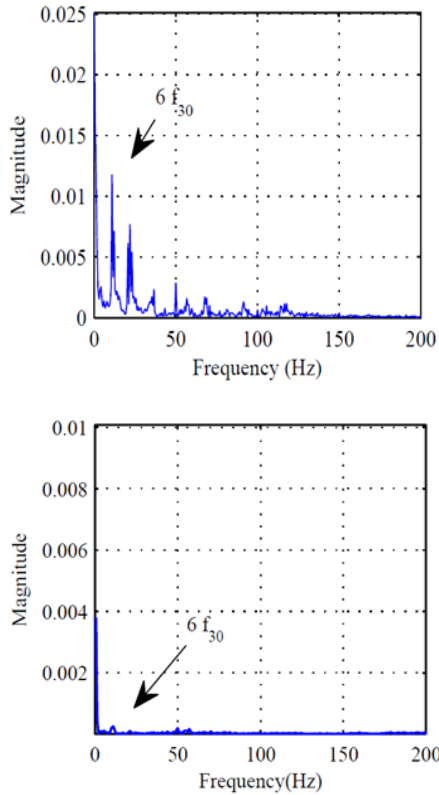


Fig. 6. Spectra measured by vibrometer without and with the proposed compensator when the speed is controlled to 30 rpm under the nominal torque 16 Nm.

V. CONCLUSION

In this paper, an active torque-harmonic compensator based on a P-Q theory has been proposed to ensure speed smoothness at low-speed high torque operation of PMSMs. The proposed approach uses PQ theory to calculate a compensation signal which is added to the main quadratic-current reference of the PMSM-controller in such a way that the undesired ripples are rejected by the PMSM-controller. The design process of the proposed compensator is detailed. It takes into account the local stability of the system and its optimal parameters are obtained by optimization. Presented experimental results demonstrated that the proposed torque-harmonic compensator can reduce the speed ripples amplitude by a ratio of two at low-speed working conditions.

This technique can be extended to other kind of machines where speed ripples are inherent like IPMSM or BLDC.

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